Digital Audio

Intro to Audio Tech 1
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Wednesday, September 10

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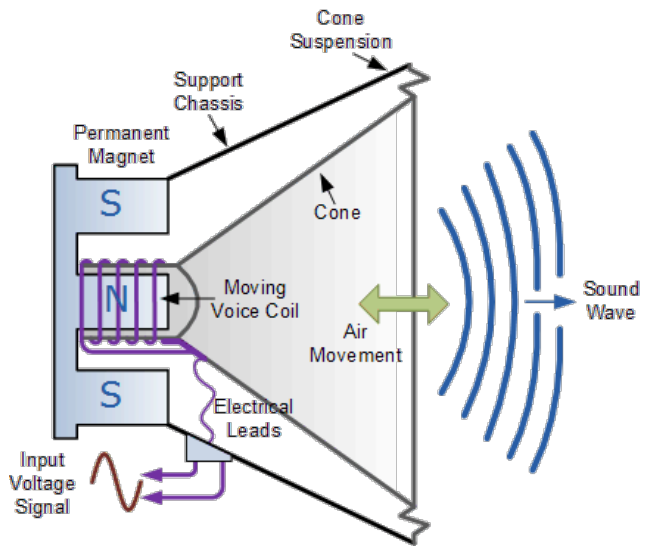
Lesson Outline

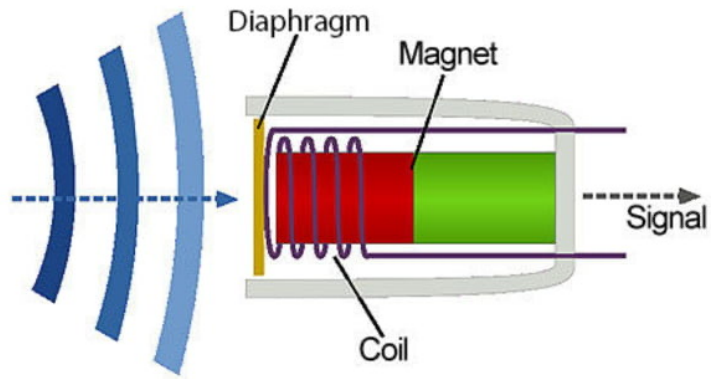
- Digital Audio
- Sampling Theorem
- Quantization
- File Formats
- Up Next…

Digital Audio

Analog

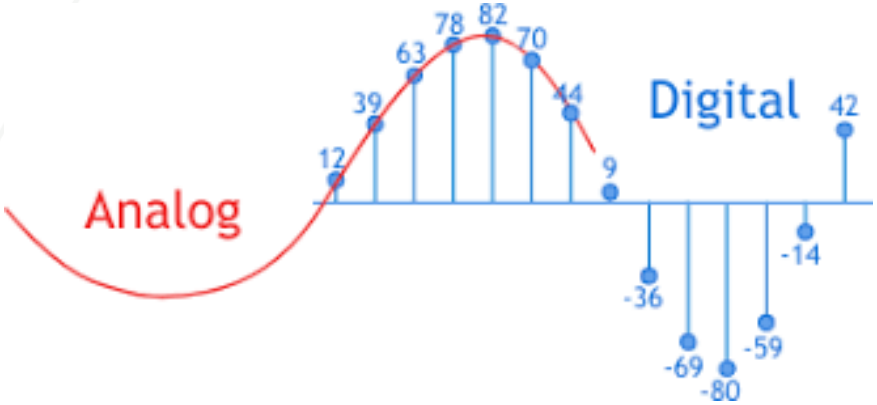
- “analogous”
- Something that emulates the changes in pressure over time
  - Microphones, speakers, wax cylinders, vinyl, tape

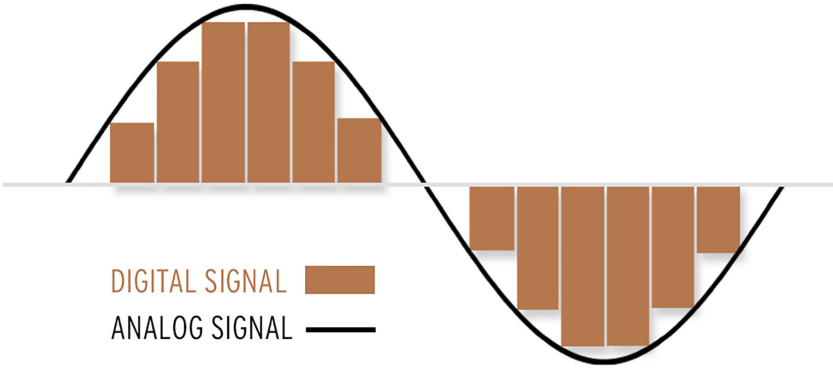




Analog to Digital

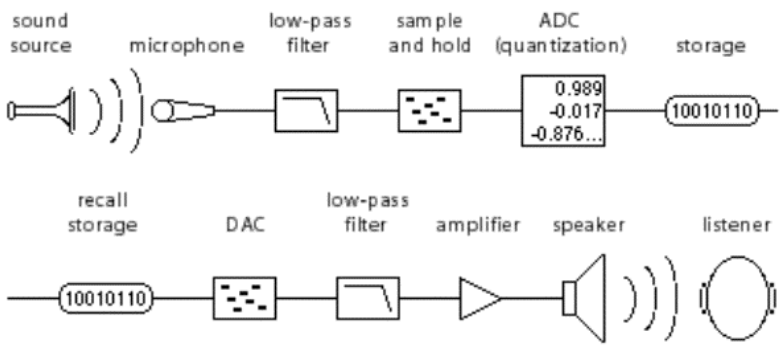
- Analog -> Continuous
- Digital -> Discrete





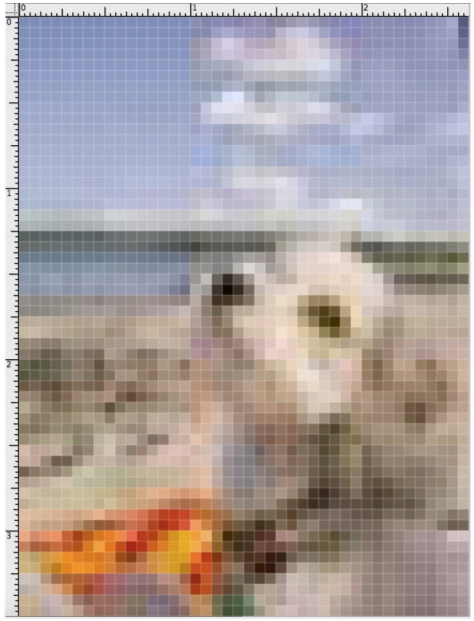
Analog to Digital

- Analog signals becoming digital through sampling and quantization
- The process of encoding waveforms in digital form is called PCM
- Analog to Digital Converter (ADC)
  - convert voltages into binary numbers at some rate
- Digital to Analog Converter (DAC)
  - Convert binary numbers into voltages at some rate (with smoothing and amplification)



Two Aspects When Digitizing a Signal

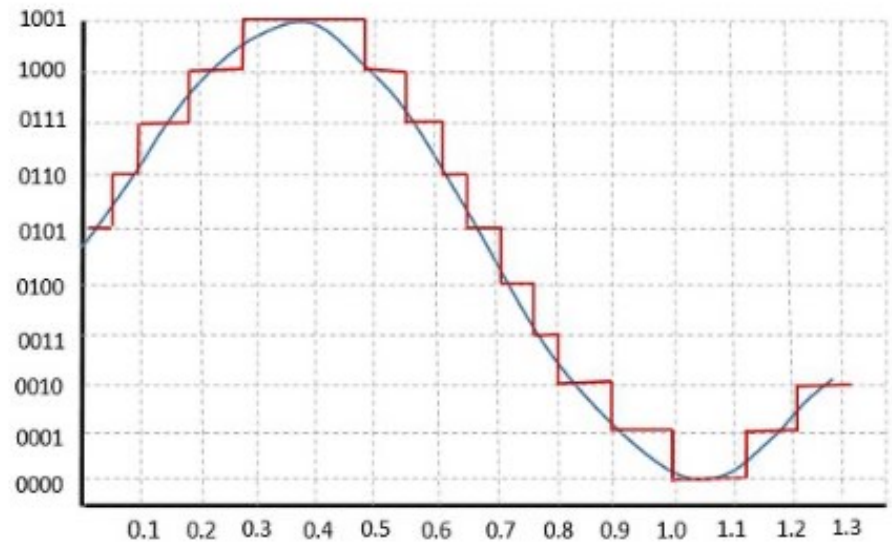
- Sampling
  - Periodically measure the analog amplitude
- Quantization
  - Round the amplitude value to a discrete value



Two Aspects When Digitizing a Signal

- Sampling
- Quantization

Approximation in Amplitude



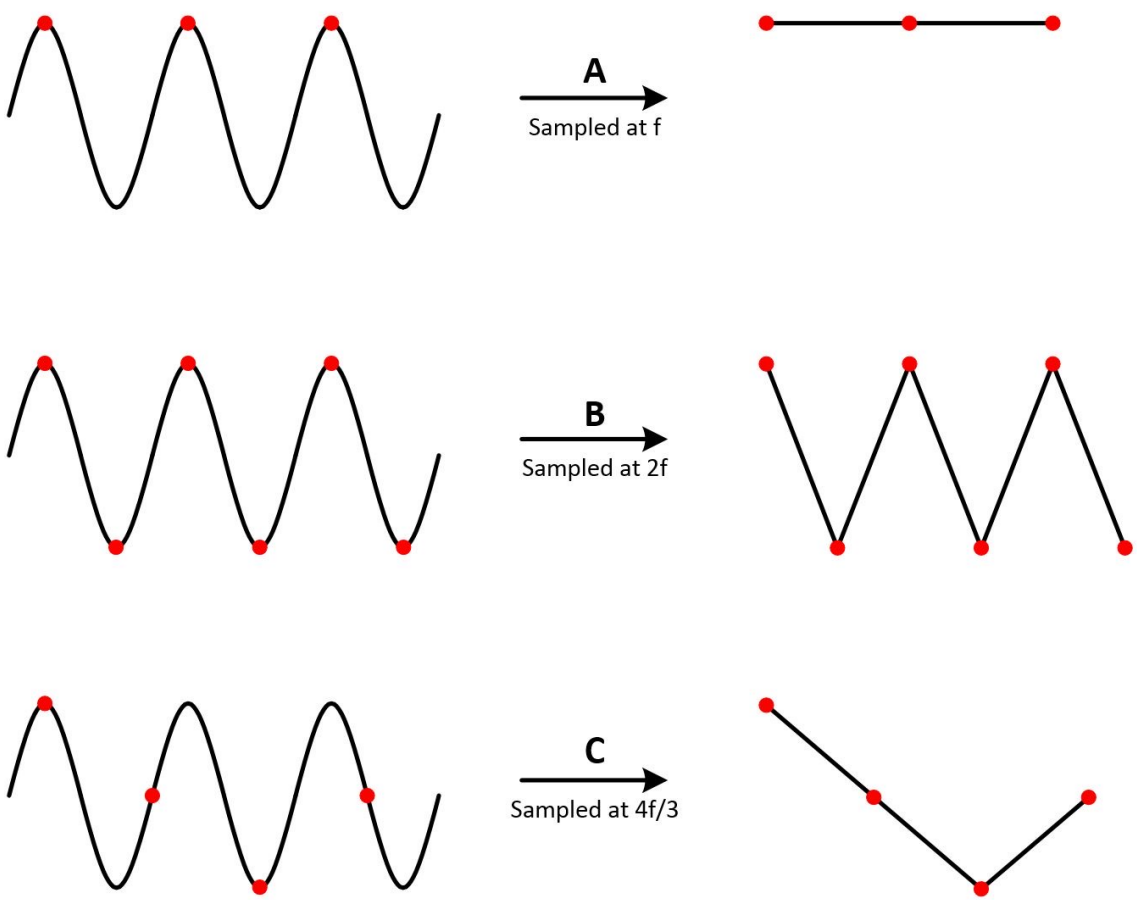
Approximation in Time

Sampling

Sampling

- Taking snapshots of the waveform at some rate
- How many times in a second should we measure the wave
- Is there information lost through sampling?
  - No (if it’s done right)

How fast should we sample?



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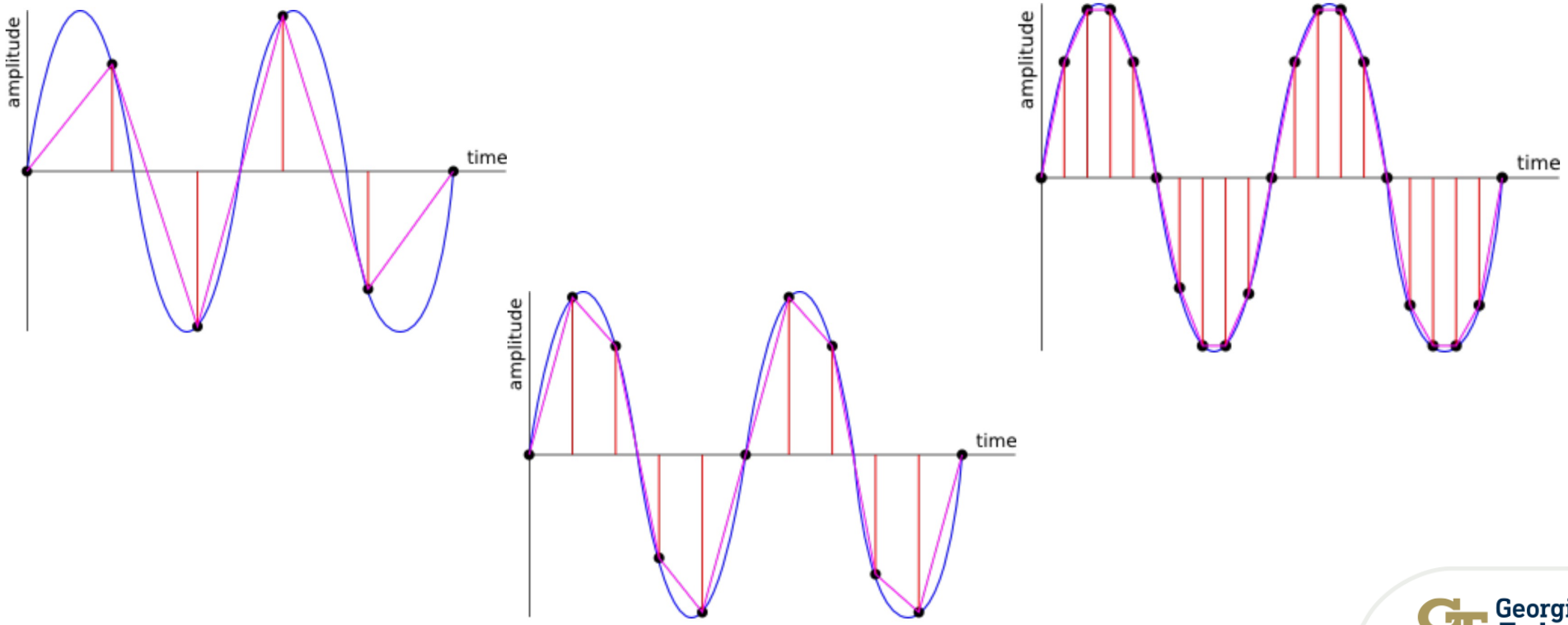

Sampling Theorem

- Nyquist Theorem
  - Theorem at Bell Labs by Harold Nyquist (1928)
  - Proof at Bell Labs by Claude Shannon (1948)

"For any given deformation of the received signal, the transmitted frequency range must be increased in direct proportion to the signaling speed. . . The conclusion is that the frequency band is directly proportional to the speed."

- The sampling frequency must be >=2x the signal frequency

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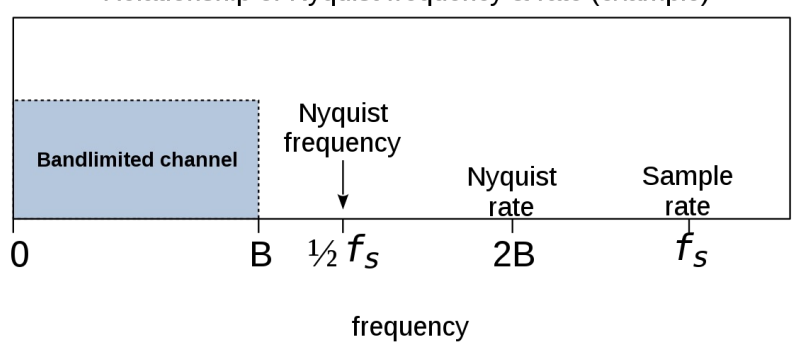
Sampling Rate

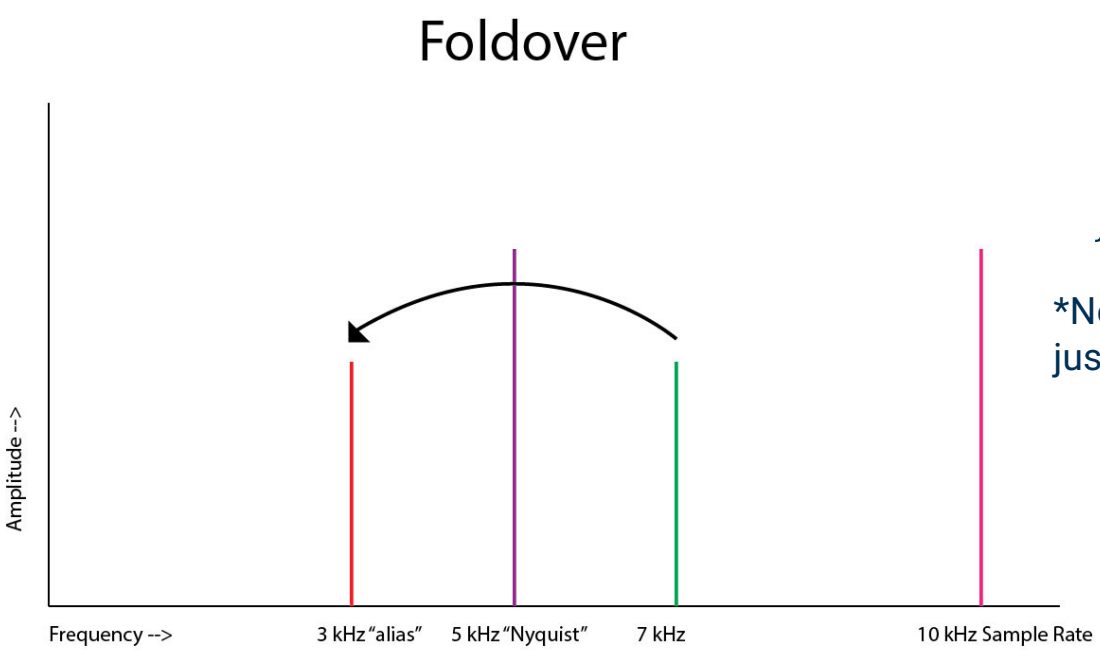
- Nyquist Theorem
  - An analog signal can be digitized without error if the sampling rate is greater than or equal to twice the highest frequency component in the analog signal
- Nyquist Rate = $2 * f_{max}$ = $2 * B$
- Nyquist Frequency = $\frac{R}{2} = \frac{SR}{2} = \frac{f_s}{2}$

Relationship of Nyquist frequency & rate (example)



Aliasing

- If we exceed the Nyquist Frequency, that high frequency will be interpreted as a lower frequency

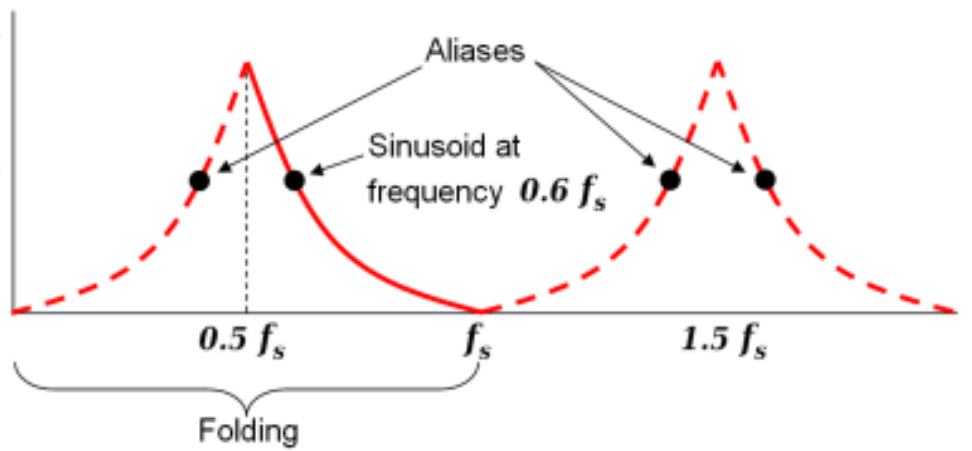


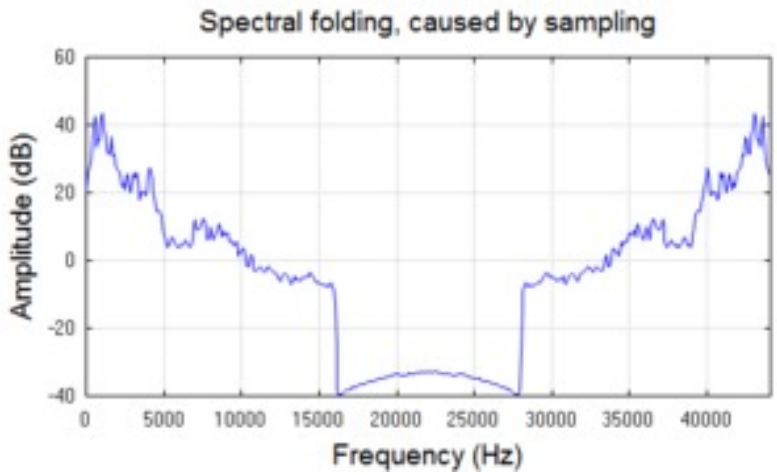
$$f_s = 10kHz \quad f = 7kHz$$
$$f_a = f - f_s = 7 - 10 = -3\ kHz$$

*Negative frequencies reflect around 0. They just mean reverse phase

Aliasing Effects

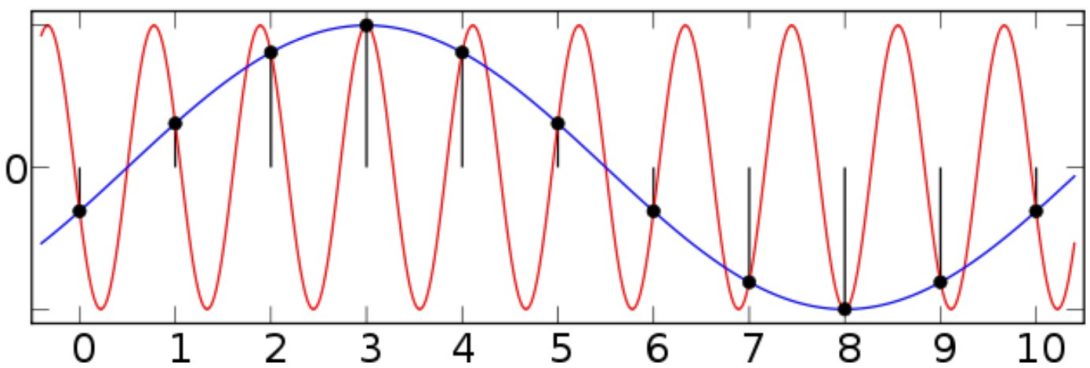
- Foldover

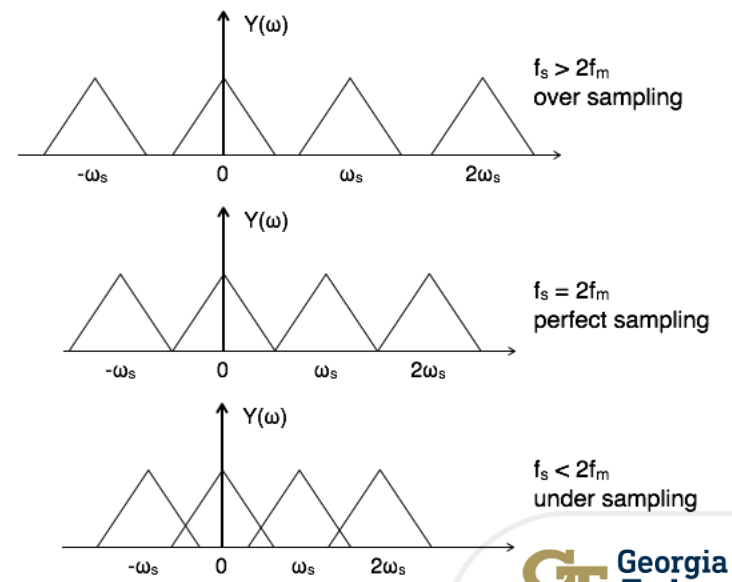




How does aliasing happen?

- Given sampling rate is f_s, our measured sinusoid is f, and Nyquist frequency is f_n. And $f < f_n$.
- We get the same sampling results for all frequencies of $f \pm$ a multiple of f_s
  - f_s = 1 Hz where f = 0.1 Hz is the same as f_s = 1 Hz where f = 0.9 Hz
  - 0.9 Hz is $f - f_s$
  - We'd also get the same at 1.1 Hz, 1.9 Hz, 2.1 Hz, 2.9 Hz





How does aliasing happen?

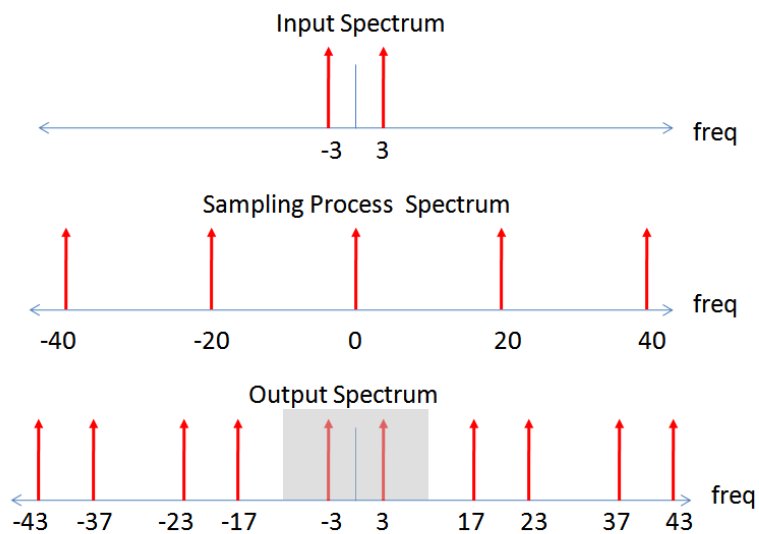
- This tells us that given sampling rate is f_s, our measured sinusoid is f, and any integer k, we cannot distinguish between the sampled values of our sinusoid f or that of another sinusoid with frequency $f \pm k * f_s$
- Another example:

f_s = 20 Hz where f = 3 Hz

$$3 \pm 1 * 20 = -17 \text{ Hz and } 23 \text{ Hz}$$

$$3 \pm 2 * 20 = -37 \text{ Hz and } 43 \text{ Hz}$$

$$3 \pm 3 * 20 = -57 \text{ Hz and } 63 \text{ Hz}$$



Calculating Output and Alias Frequencies

- We can use this pattern to calculate the digitized frequency given a sampling rate and measured frequency

$$f_o = f - N * f_s$$

where N is the whole number multiple of f_s that's closest to f

$$N = round(\frac{f}{f_s})$$

Practice

- Given $f_s = 10000$, what is the Nyquist frequency and what is the output for the following frequencies?
  - $f = 3000$
    - Nyquist is 5000, so output is 3000
  - $f = 6000$
    - $f_o = f - N * f_s = 6000 - \frac{6000}{10000} * 10000 = 6000 - 1 * 10000 = -4000$
  - $f = 13000$
    - $f_o = f - N * f_s = 13000 - \frac{13000}{10000} * 10000 = 13000 - 1 * 10000 = 3000$
  - $f = 21000$
    - $f_o = f - N * f_s = 21000 - \frac{21000}{10000} * 10000 = 21000 - 2 * 10000 = 1000$

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Sampling Rates

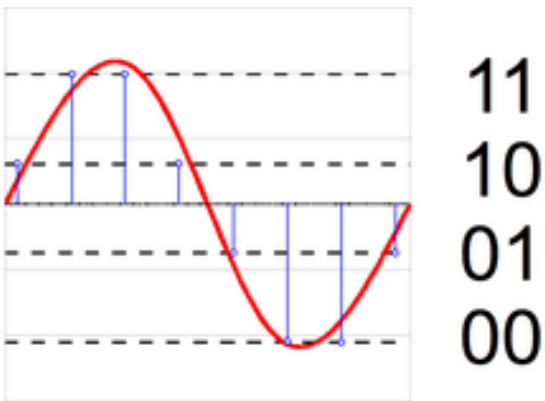
- 44.1 kHz
  - That means 44,100 numbers are stored every second
- Why 44.1?
  - Human hearing is 20 Hz to 20 kHz
  - Why not just 40 kHz?
  - Easier filtering for aliasing at the 22.05 kHz mark

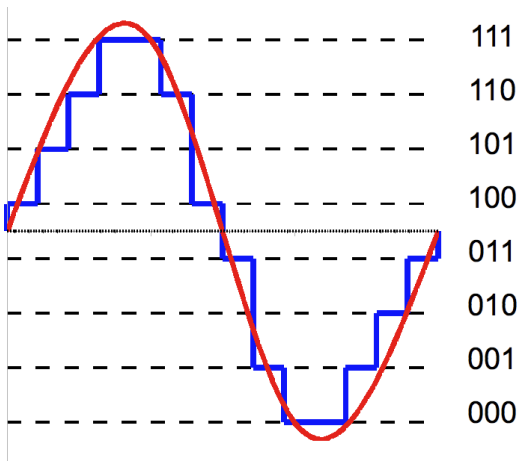
System	Standard audio sampling rates (Fs)
Telephone	8 kS/s
Compact disc	44.1 kS/s
Professional audio (DAT)	48 kS/s
DVD (6 channel audio channel)	96 kS/s
A-DVD, Blu-ray disc	192 kS/s

Quantization

Quantization

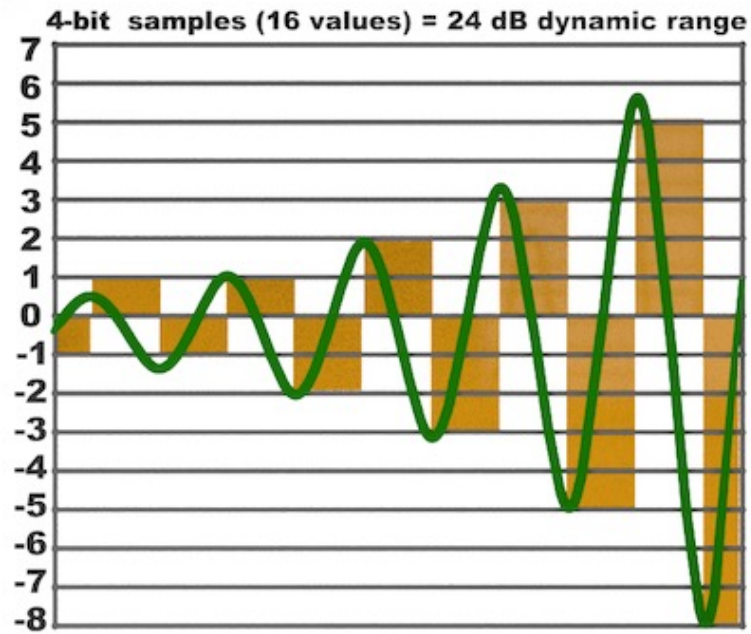
- How many different amplitude values can we store?
- Depends on our bit depth
  - How many bits can we use to represent our numbers
- Bit depth = 2^n where n is the number of bits
  - 8 bit has 256 levels
  - 16 bit has 65,536 levels





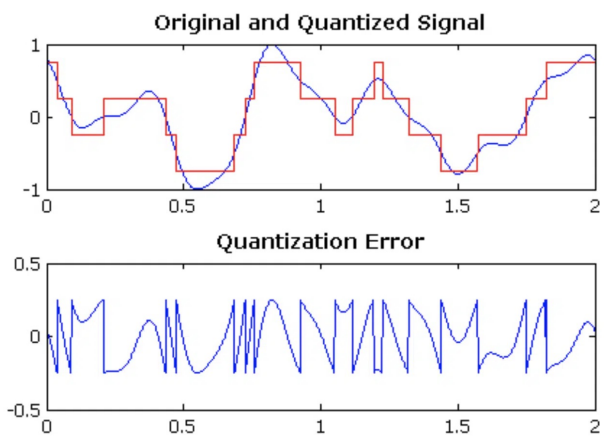
Quantization – Dynamic Range

- Dynamic range is the amount of loudness you can capture
- Roughly - number of bits x 6 dB
  - (for linear PCM encoding)
  - 8-bit upper limit is 48dB
  - 16-bit upper limit is 96dB
  - 20-bit upper limit is 120dB
- We can change encodings to try to use less bits



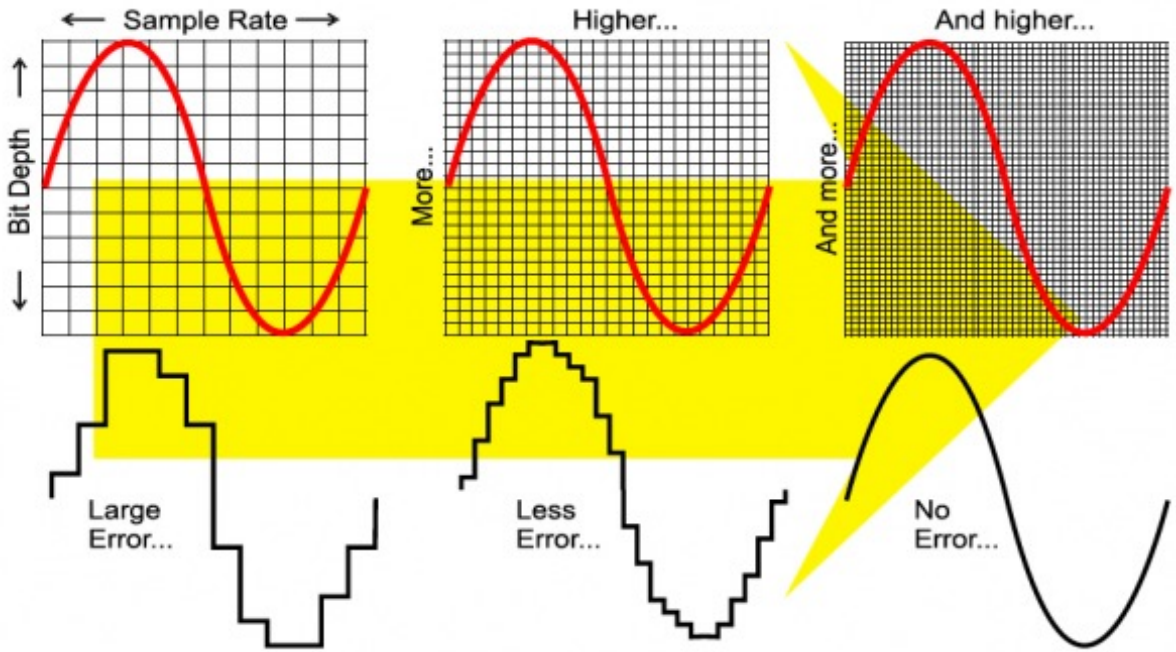
Quantization

- Quantization error
- Because we are rounding, quantization adds noise
  - The difference between the actual analog signal and the rounded signal
- The SNR caused in linear quantization is reduced by around 6dB
  - 16 bit = 96 dB, 8 bit = 48 dB
- Less dynamic range
  - Our softs and louds are closer together with less bits



How everything works together

- Bit Rate = bits per second = bits per sample * samples per second
  - Stereo CD quality: 16 bits * 44.1 kHz * 2 channels = 1411.2 Kbps



File Formats

Calculating Storage Space

- If we have 1 minute of stereo audio at 16-bit, 44.1kHz

60 seconds x 44,100 samples/sec x 2 channels = **84 million bits**

divide by 8 bytes

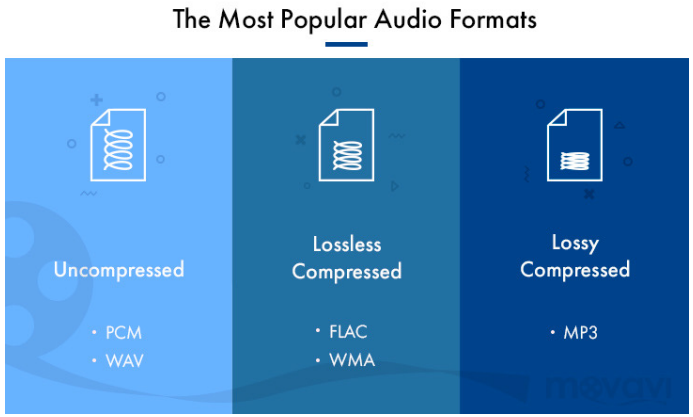
Divide by 1024 kB

Divide by 1024 MB

= ~10 MB

Lossless vs Lossy File Formats

- Data compression
- Lossless
  - Preserves 100% of the information in the raw audio stream
- Lossy
  - Eliminate data that doesn't impact the sound (ideally)
  - Information is thrown away



Uncompressed

- WAV and AIFF
  - Windows vs Apple devs
- CD quality audio – 16 bit, 44.1kHz
  - Supports up to 32-bit, 192kHz
- Use PCM, pulse code modulation
- AIFF also supports metadata tagging

- Very large files

Lossless

- FLAC
  - Free Lossless Audio Codec
- Supports up to 32-bit, 192kHz
- ALAC (Apple Lossless) and WMA (Windows Media Audio)
  - .m4a extension

- Lossless audio without massive files

Lossy

- MP3
  - MPEG-1 Audio Layer 3
  - Encoded at 128kbps, up to 320 kbps
  - Up to 48 kHz
- AAC
  - Advanced Audio Coding
  - Developed with a team of tech companies
  - Encoded at 256kbps
  - Up to 96kHz
  - Better quality than MP3

- Compact storage but at a cost

Looking at Digitized Sound in Max

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Up Next…

- Readings!
- Assignment 1 is due Friday at 11:59pm
- Assignment 2 has been posted due Sept 26th
- Exam 1 is on Sept 24
  - Study guide will be posted by Sunday this week